

Real-Time Communication using Django Channels

Project: Django Channels Chat

Module: WebSocket & Real-Time Communication

Prepared By: _____

Project Overview

This report explains the implementation of the real-time communication module using Django Channels, WebSockets, Redis and Daphne.

Objectives

- Configure Django Channels
- Enable WebSocket communication
- Broadcast messages in real time
- Support multiple chat rooms
- Implement join/leave notifications

Technologies Used

Django 6

Django Channels

Redis

Daphne

Docker

JavaScript

Architecture

Browser ↔ WebSocket ↔ Django Channels ↔ Redis Channel Layer ↔ Other Browsers

Project Structure

chat_project/

chat/

consumers.py

routing.py

views.py

templates/chat/home.html

chat_project/
asgi.py
settings.py
manage.py

Workflow

1. User joins a room.
2. WebSocket connection opens.
3. Consumer accepts the connection.
4. User joins a Redis group.
5. Messages are broadcast.
6. All users receive updates instantly.

Files Explained

settings.py: Channels + Redis
asgi.py: Protocol routing
routing.py: WebSocket URL mapping
consumers.py: WebSocket logic
home.html: WebSocket client

Features

Dynamic rooms
Real-time messaging
JSON communication
Join notifications
Leave notifications
Multi-user chat

Testing

Run Redis:
docker start redis-chat

Run Daphne:
python -m daphne chat_project.asgi:application

Open two browser windows and verify messaging.

Conclusion

The real-time communication module was successfully implemented and is ready for backend integration.

Screenshots to Add

- Home page
- Two users chatting
- Daphne terminal
- Docker Redis running
- Join notification
- Leave notification



Django Channels Chat

Join a Room

Messages

```
(venv) PS C:\Users\Sunny\Desktop\projectsss\django_channels_practice> docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED    STATUS    PORTS                               NAMES
27e28047badd   redis:7    "docker-entrypoint.s..." 40 minutes ago    Up 40 minutes    0.0.0.0:6379->6379/tcp, [::]:6379->6379/tcp    redis-chat
(venv) PS C:\Users\Sunny\Desktop\projectsss\django_channels_practice> python -m daphne chat_project.asgi:application
ASGI FILE LOADED
Open Website
```

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127.0.0.1:8000

Django Channels Chat

Join a Room

Join Room

Send

Messages

Connected to room: general

- bunny joined the room.
- sunny: hello
- bunny: hi

Two Users Chatting

Django Channels Chat

Join a Room

Messages

Connected to room: general

 sunny joined the room.

 bunny joined the room.

sunny: hello

bunny: hi

 bunny left the room.

Django Channels Chat

Join a Room

Messages

Connected to room: general

 bunny joined the room.

sunny: hello

bunny: hi

Connected to room: sports

 bunny joined the room.

Connected to room: sports

 bunny joined the room.

Leave Notification

Daphne Server Running

Docker Redis Running

```
(venv) PS C:\Users\Sunny\Desktop\projectsss\django_channels_practice> docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                               NAMES
27e28047badd   redis:7    "docker-entrypoint.s..." 40 minutes ago Up 40 minutes 0.0.0.0:6379->6379/tcp, [::]:6379->6379/tcp  redis-chat
(venv) PS C:\Users\Sunny\Desktop\projectsss\django_channels_practice> python -m daphne chat_project.asgi:application
ASGI FILE LOADED
```

Open Website

Ln 161, Col 8 Spaces: 4 UTF-8 CRLF {} HTML BLACKBG



127.0.0.1:8000

Django Channels Chat

Join a Room

Messages

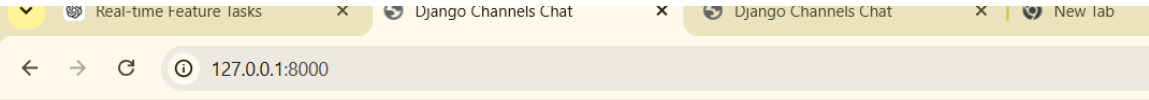
Connected to room: general

 sunny joined the room.

 bunny joined the room.

sunny: hello

bunny: hi



Django Channels Chat

Join a Room

Messages

Connected to room: general

- Someone joined the room.
- Someone joined the room.

sunny: hello